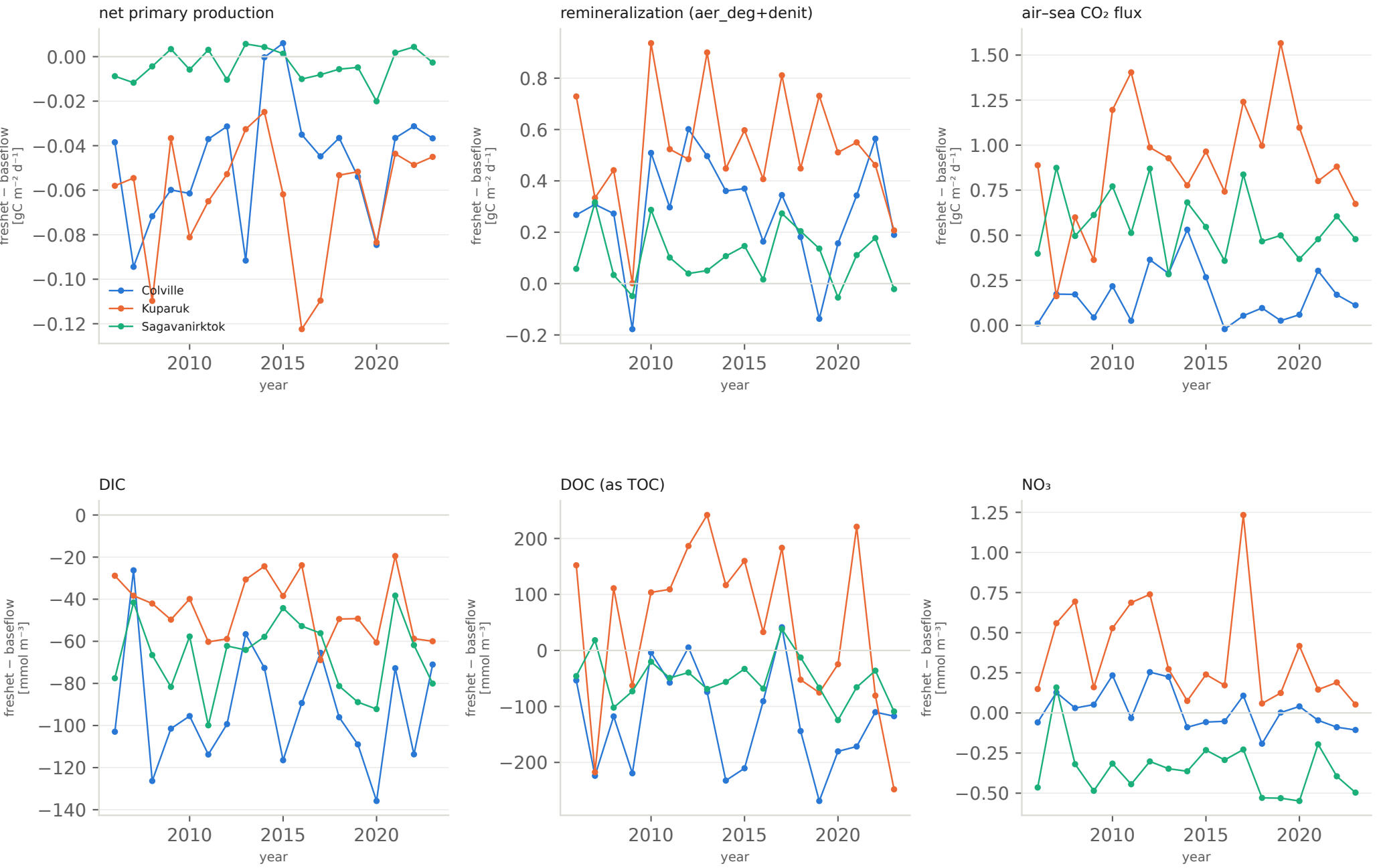
Freshet-window vs. baseflow-window biogeochemistry, 2006–2023

Per calendar year: mean value on freshet days (Q > 3× that year's mean) minus the mean on all other days. Positive = elevated during the freshet.



New calculation (not a trapezoidal annual budget -- freshet days are non-contiguous, so a per-day-subset mean is used instead). Uses runs/interannual_full/. Upstream (river) boundaries are CONSTANT for every solute except T and TOC -- same limitation as the original 44-yr record. Marine (mouth) boundaries, by contrast, are genuinely time-varying here for EVERY solute (S, DIC, ALK, NO3, NH4, PO4, dSi, O2, TOC -- ECCO-Darwin v5, 2005-2023), which is new relative to the original figures (only T was marine-time-varying there). Structure seen here reflects BOTH the model's own internal dynamics acting on a constant upstream source AND a genuinely time-varying ocean signal propagating in from the mouth. See CLAUDE.md -> "Interannual forcing" -> "Full-forcing interannual (2005-2023)".